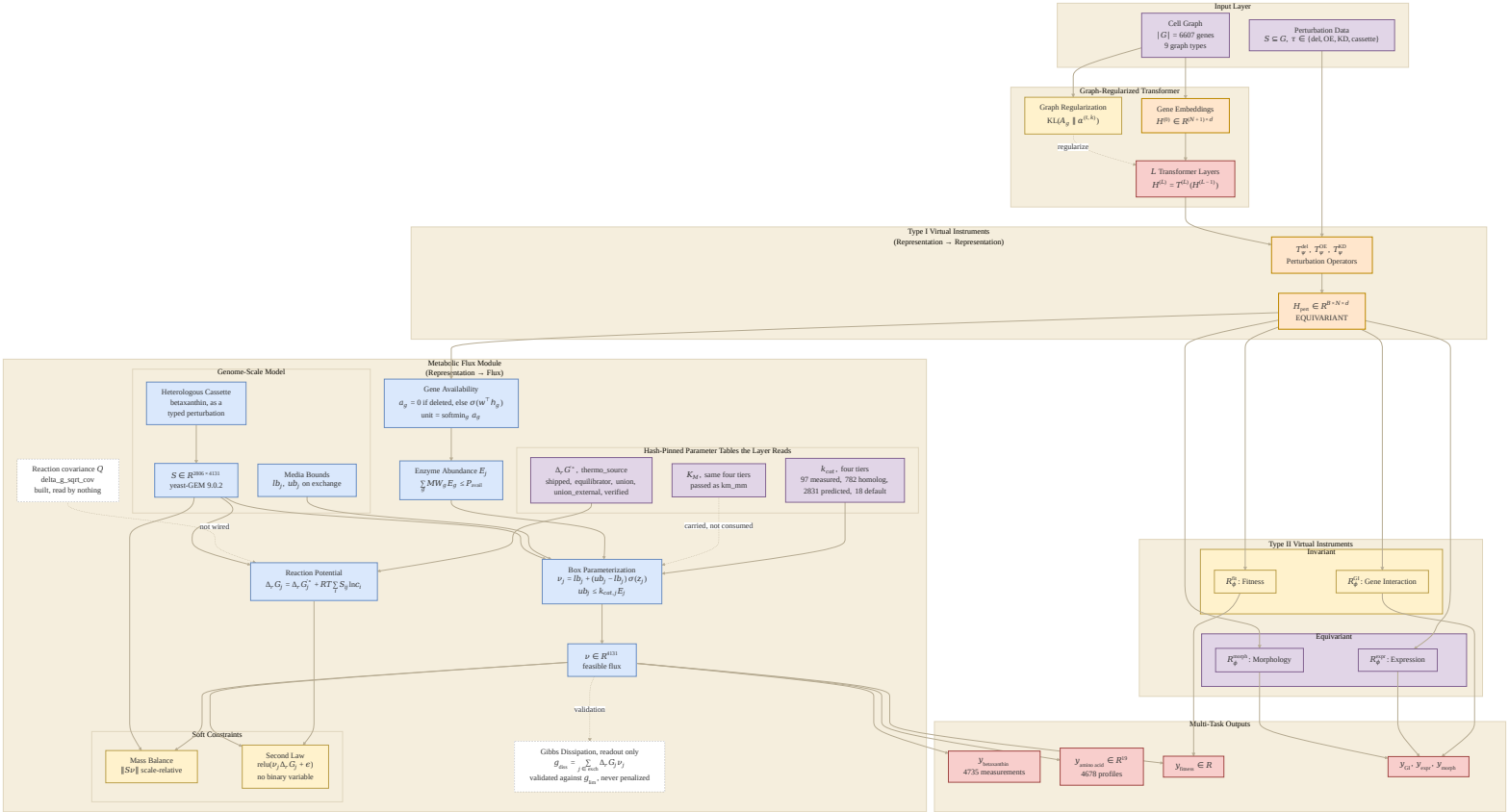

Input Layer
Cell Graph
$|G| = 6607$ genes
9 graph types
Perturbation Data
$S \subseteq G,\ \tau \in$ {del, OE, KD, cassette}
Graph-Regularized Transformer
Graph Regularization
$\mathrm{KL}(A_\theta \,\|\, \alpha^{(L,K)})$
Gene Embeddings
$H^{(0)} \in R^{N+1)\times d}$
regularize
L Transformer Layers
$H^{(L)} = T^{(L)}(H^{(L-1)})$
Type I Virtual Instruments
(Representation → Representation)
$T_\theta^{\mathrm{del}},\ T_\theta^{\mathrm{OE}},\ T_\theta^{\mathrm{KD}}$
Perturbation Operators
$H_{\mathrm{pert}} \in R^{B\times N\times d}$
EQUIVARIANT
Metabolic Flux Module
(Representation → Flux)
Genome-Scale Model
Heterologous Cassette
betaxanthin, as a
typed perturbation
$S \in R^{2806\times 4131}$
yeast-GEM 9.0.2
Media Bounds
$lb_j,\ ub_j$ on exchange
Reaction covariance Q
delta_g_sqrt_cov
built, read by nothing
not wired
Gene Availability
$a_g = 0$ if deleted, else $\sigma(w^\top h_g)$
unit = softmin$_g$ a_g
Enzyme Abundance E_j
$\sum_g MW_g E_g \le P_{\mathrm{avail}}$
Hash-Pinned Parameter Tables the Layer Reads
$\Delta_r G'^\circ$, thermo_source
shipped, equilibrator, union,
union_external, verified
K_M, same four tiers
passed as km_mm
k_{cat}, four tiers
97 measured, 782 homolog,
2831 predicted, 18 default
carried, not consumed
Reaction Potential
$\Delta_r G_i = \Delta_r G_i'^\circ + RT\sum_\tau S_{i\tau} \ln c_\tau$
Box Parameterization
$\nu_j = lb_j + (ub_j - lb_j)\,\sigma(z_j)$
$ub_j \le k_{\mathrm{cat},j} E_j$
$\nu \in R^{4131}$
feasible flux
validation
Soft Constraints
Mass Balance
$\|S\nu\|$ scale-relative
Second Law
relu$(\nu_j \Delta_r G_j + \epsilon)$
no binary variable
Gibbs Dissipation, readout only
$g_{\mathrm{diss}} = \sum_{j\in\mathrm{each}} \Delta_r G_j \nu_j$
validated against g_{ext}, never penalized
Type II Virtual Instruments
Invariant
R_θ^{fit} : Fitness
R_θ^{GI} : Gene Interaction
Equivariant
$R_\theta^{\mathrm{morph}}$: Morphology
R_θ^{expr} : Expression
Multi-Task Outputs
$y_{\text{betaxanthin}}$
4735 measurements
$y_{\text{amino acid}} \in R^{19}$
4678 profiles
$y_{\mathrm{fitness}} \in R$
y_{GI}, y_{expr}, y_{morph}